Final Test Coverage Report

DDP-087 • Coverage evidence and verification assessment

Darshana Opticals — Optical Shop Management System (OSMS)

DDP-087 | Evidence date: 25 October 2026 | Document status: Evidence-backed draft pending final SRS/CI traceability

Document control: Prepared from locally executed Jest/Vitest logs and DDP-087 issue requirements. No manual or E2E outcomes are invented.

1. Executive summary

Two successful local automated test executions were independently recorded: 2,075/2,075 backend tests and 1,341/1,341 frontend tests, totaling 3,416 passing tests across 228 test files/suites. Both measured repositories exceed 80% in each overall code-coverage metric. This is evidence of automated suite health at the tested local revisions, not proof that every approved SRS requirement, API endpoint, release commit, or hosted workflow has been fully verified.

Measure	Backend (Jest)	Frontend (Vitest)
Passing tests	2,075 / 2,075	1,341 / 1,341
Passing suites/files	118 / 118	110 / 110
Failed tests	0	0
Statements	92.34%	88.05%
Branches	81.68%	81.80%
Functions	97.86%	90.95%
Lines	93.00%	88.86%

2. Scope, baseline and evidence provenance

In scope: local Jest backend unit/integration tests, local Vitest/React Testing Library frontend unit/component tests, associated instrumentation and measured code coverage, plus a traceability framework for integration, browser E2E, security, performance, and requirements testing. Out of scope for these two execution logs: independently validated production readiness, hosted payment settlement, exhaustive manual UAT, and full API-to-test or SRS-to-test mapping.

Item	Backend	Frontend
Repository	darshana-opticals-osms/backend	darshana-opticals-osms/frontend
Local tested SHA (previously collected)	47eb042 (reconfirm with git rev-parse HEAD)	3e6fa9b (reconfirm with git rev-parse HEAD)
Local branch (previously collected)	docs/91-post-deployment-security-verification	test/91-owasp-security-verification
Framework	Jest; Supertest where used	Vitest 5.0.5; React Testing Library
Test execution date	25 October 2026	25 October 2026
Recorded duration	265.944 seconds	78.23 seconds
Result artifact	backend/coverage/coverage-summary.json	frontend/coverage/coverage-summary.json

Item	Backend	Frontend

Revision note: the SHAs above came from earlier local branch inspection rather than being printed inside the uploaded final coverage logs. Confirm them against git rev-parse HEAD before labeling the documents as final release evidence. The logs themselves establish the test counts and percentages.

3. Reproduction and environment

Backend command: npx jest --runInBand --testTimeout=30000 --coverage --coverageReporters=json-summary --coverageReporters=text-summary. MongoDB Memory Server used the installed MongoDB 8.3.11 executable through session-only MONGOMS_SYSTEM_BINARY; SYSTEM_BINARY_VERSION_CHECK=false and RUNTIME_DOWNLOAD=false. A separate 3/3 database foundation test passed before the full run. The earlier failed run was affected by a stuck MongoDB 8.2.6 binary download; it must not be counted as code regression evidence.

Frontend command: npm run test:coverage -- --coverage.reporter=text --coverage.reporter=json-summary. V8 was the reported coverage provider. Frontend console emitted non-failing React Router future-flag warnings and React act(...) warnings. Record these as maintenance/testing hygiene observations.

4. Coverage target assessment

Guideline requirement	Evidence	Assessment
80% code coverage for new code	All four measured overall metrics >80% in each repository	Overall threshold exceeded; new-code-only coverage not separately measured
90% branch coverage for critical business logic	Overall branch coverage: backend 81.68%; frontend 81.80%	NOT VERIFIED by aggregate numbers; per-critical-module review needed
At least one integration test per API endpoint	Integration tests present; full route inventory/test mapping unavailable	NOT VERIFIED
Critical end-to-end workflows	DDP-084 provides separate execution evidence	Must cross-check actual scenarios, URLs and statuses
Final SRS requirement traceability	Approved SRS requirement inventory not included in execution logs	NOT VERIFIED

5. Backend automated test coverage

Jest reported 118 passed suites and 2,075 passed tests. Overall statement coverage: 6,800/7,364; branch coverage: 4,958/6,070; function coverage: 778/795; line coverage: 6,571/7,065. The additional JSON field branchesTrue=100% is not equivalent to ordinary branch coverage and is not substituted for 81.68%.

Representative observed passing backend test suites include: database foundation; health; admin staff management; staff/doctor lifecycle security; chatbot session API; inventory image lifecycle/upload; loyalty read API; password reset service; prescription service; payment validation/model/monitoring; audit services; reports; authorization middleware; orders, checkout and inventory services. The list establishes observed suite-level execution, not endpoint-by-endpoint integration coverage.

6. Frontend automated test coverage

Vitest reported 110 passing test files, 1,341 passing tests, no failures. Overall coverage is 88.05% statements, 81.80% branches, 90.95% functions and 88.86% lines.

Representative executed areas: sign-up, login/password recovery, protected routes and role navigation; customer profile; product search and detail; cart/checkout/payment entry and result; order history and staff order controls; appointments/doctor sessions; prescriptions; inventory/reorder alerts; loyalty; chatbot; sales/inventory reporting; administrative staff/doctor management. Most frontend tests mock API behavior and should not be conflated with full-stack E2E validation.

7. Module-level risk observations

Frontend module	Observed branch %	Recommended interpretation
Appointment booking page	71.55%	Add boundary/error-state tests if considered critical
Payment entry page	68.70%	Critical payment initiation paths need branch review
Checkout entry page	68.05%	Checkout handling warrants targeted test review
Admin staff editor	66.37%	Sensitive role/reassignment flows warrant review
Staff move appointment page	57.64%	Uncovered move/conflict paths warrant review
Protected routes	100.00%	Strong module result; does not replace server authorization
Payment return page	100.00%	Tested component result only; hosted gateway not proven

These are file-level branch percentages from the frontend console report. The designation “critical” must be finalized by the team against SRS risk categories; no 90% compliance claim is made for them. Backend critical-file percentages require its detailed JSON file, not only the aggregate summary.

8. API integration coverage assessment

The backend suite contains integration tests, but the 118/118 result alone cannot establish one test per distinct HTTP endpoint. The required API register must be generated from the implemented router definitions, then paired with matching integration test names and methods. Authentication, permitted roles, negative-access checks and execution status must be confirmed per route. Until that comparison is completed, endpoint coverage is “Not verified.”

Endpoint group	Observed test evidence	Full endpoint inventory
Authentication/profile/recovery	Jest authentication and password-reset; frontend auth flows	Pending route-by-route review
Inventory/orders/payments	Jest inventory/checkout/payment; frontend order and payment tests	Pending route-by-route review
Appointments/prescriptions	Jest appointment/prescription; frontend page flows	Pending route-by-route review
Reports/loyalty/chatbot	Jest reporting/loyalty/chatbot; frontend services and flows	Pending route-by-route review
Admin/audit/notifications	Jest admin/audit/notification; frontend admin flows	Pending route-by-route review

9. E2E, security, performance and defect evidence

DDP-084 (infra issue #11 / merged PR #19) is the source for browser-level integration and end-to-end evidence. Prior project review referenced 10 individually reported Playwright checks, including 7/7 later repeated authorization checks; retain the actual logs and exact environment before promoting these to a final verified matrix. Hosted PayHere callback/payment verification was reported as constrained by a merchant-configuration error; do not label a complete hosted purchase journey “Pass” without new proof.

DDP-085 security: reference the existing documents/security/owasp-checklist.pdf and backend issue #91, including the precise tested conditions, scope and any outstanding limitations. DDP-086 performance: reference documents/testing/performance-report.pdf and backend issue #92; workload, p95 latency, error rate and configured test environment must be copied from that measured report rather than inferred. DDP-092 defect closure: reference infra issue #14 and associated regression evidence; merged status alone is not equivalent to verified zero residual defects.

10. CI and release traceability

CI workflow run IDs, artifact URLs and last release SHAs are not present in the uploaded coverage logs. Record exact GitHub Actions run URLs and confirm the tested commits match the delivered release. If the local SHAs differ from final develop/main, state the difference and obtain appropriate release verification rather than silently treating local results as production sign-off.

11. Open gaps and corrective follow-up

- Confirm actual tested SHAs, target branch and release commit; attach persistent log/coverage artifacts.
- Map every SRS functional and applicable nonfunctional requirement to test cases; do not invent SRS IDs.
- Inventory every implemented backend API endpoint and document at least one matching integration test or explicit gap.
- Assess each team-designated critical business-logic module against its 90% branch target.
- Cross-check DDP-084 browser E2E and hosted payment outcome with logs; mark unresolved items Blocked/Not Verified.
- Import actual DDP-085, DDP-086 and DDP-092 measured results and traceable issues/CI evidence.

12. Conclusion and sign-off status

Conclusion: the reported backend and frontend local automated suites PASSED (3,416/3,416 tests), and both independently measured aggregate code-coverage profiles are above 80% in statements, branches, functions and lines. Separate proof remains necessary for critical-module branch targets, route completeness, full SRS traceability, E2E hosted payments and final release parity. Therefore this report supports automated-test completion at the observed local revisions, but is not an unconditional final UAT or production-readiness certificate.

Prepared for DDP-087 review. QA reviewer: _____ Review date: _____ Release SHA(s): _____

Appendix A — SRS requirement alignment

SRS	Business capability	Representative test evidence	Boundary / qualification

FR-001	Authentication and user roles	Jest auth; frontend Login/Signup, route guards	Automated evidence; hosted full-scope not established
FR-002	Customer profile / prescription history	ProfilePage; PrescriptionHistoryPage	Automated evidence
FR-003	Catalogue	ProductsPage; productService	Partial implementation: style filter gap
FR-004	Cart / checkout	CartPage; CheckoutEntryPage; checkout.service	Automated evidence
FR-005	Inventory	InventoryPage; inventory integration tests	Partial implementation: no verified real-time sync
FR-006	Reorder alerts	InventoryReorderAlerts; reorderAlert.service	Automated evidence
FR-007	Doctor sessions / appointment	AppointmentBookingPage; appointment.service	Modified by ADR-007
FR-008	Appointment communication	ADR-009	Approved manual process; automated reminders not in scope
FR-009	PayHere	PaymentEntryPage; payment unit tests	Hosted payment blocked/not verified
FR-010	Loyalty	loyaltyReadApi; CustomerLoyaltyPage	Accrual tested; redemption out of scope
FR-011	Chatbot	chatbotSessionApi; ChatbotFlow	Automated evidence
FR-012	Reports	SalesReportPage; InventoryReportPage	Automated evidence
FR-013	Prescriptions	prescription.service; PrescriptionFlows	Automated evidence
FR-014	Order statuses	OrderDetailsPage; StaffOrderStatusControl	Modified by ADR-004; no courier sync

Appendix B — Source-reviewed API route inventory

The backend source inventory identifies 73 route declarations across 17 route groups. The detailed method, path, access context and possible related test evidence are included in the companion Test Case Register Appendix C. This is an assessment of implementation route declarations only. At least one executed integration assertion per endpoint has NOT been demonstrated for all routes; do not label endpoint coverage complete.

Backend area	Routes inventoried	Endpoint assertion status
auth	4	Not independently verified per endpoint
admin	10	Not independently verified per endpoint
profile	2	Not independently verified per endpoint
product	2	Not independently verified per endpoint
inventory	10	Not independently verified per endpoint
prescription	3	Not independently verified per endpoint
branch	1	Not independently verified per endpoint
cart	4	Not independently verified per endpoint
checkout	1	Not independently verified per endpoint
order	5	Not independently verified per endpoint
payment	9	Not independently verified per endpoint
appointment	10	Not independently verified per endpoint
doctor	2	Not independently verified per endpoint
audit	1	Not independently verified per endpoint

loyalty	2	Not independently verified per endpoint
report	6	Not independently verified per endpoint
health	1	Not independently verified per endpoint

Appendix C — Critical business logic branch assessment

Historical aggregate branch coverage: backend 81.68%; frontend 81.80%. Source log provides frontend file-level data, but no backend per-file coverage table. A 90% branch threshold for critical business modules therefore cannot be certified as achieved from these records.

Selected risk-sensitive module	Measured branch coverage	Assessment
Backend checkout/order/payment/inventory/appointment/authorization	Backend per-file branch metrics unavailable in uploaded text summary	Not verified; export per-file JSON for critical modules
Frontend PaymentEntryPage.jsx	68.70%	Below 90%
Frontend CheckoutEntryPage.jsx	68.05%	Below 90%
Frontend AppointmentBookingPage.jsx	71.55%	Below 90%
Frontend AdminStaffEditor.jsx	66.37%	Below 90%
Frontend CartProvider.jsx	68.08%	Below 90%
Frontend StaffMoveAppointmentPage.jsx	57.64%	Below 90%
Frontend InventoryEditor.jsx	92.40%	At/above 90% for this file
Frontend orderService.js	100%	At/above 90% for this file
Frontend paymentService.js	86.20%	Below 90%

Module criticality requires confirmation by the team. Low coverage above is not automatically a defect but is evidence that full 90% per-critical-module compliance cannot be claimed. Follow-up should add targeted tests to uncovered branches; no production refactoring is requested by this documentation review.

Appendix D — QA execution, CI, and provenance

Backend historical local outcome: 118 of 118 suites, 2,075 of 2,075 tests passing; Jest coverage run duration 265.944 seconds. Frontend historical local outcome: 110 of 110 files, 1,341 of 1,341 tests passing; Vitest 5.0.5 duration 78.23 seconds. Both runs captured on 25 October 2026. Earlier local branch observations: backend 47eb042, frontend 3e6fa9b; uploaded runner output does not prove these SHA values, and they MUST be rechecked before asserting exact baseline.

Backend command: npx jest --runInBand --testTimeout=30000 --coverage --coverageReporters=json-summary --coverageReporters=text-summary. Frontend command: npm run test:coverage -- --coverage.reporter=text --coverage.reporter=json-summary. CI run URLs not present in supplied runner output; no URL fabricated. These are local run measurements, NOT demonstrated release-SHA CI execution. GitHub Actions run IDs, timestamp, branch, SHAs, artifacts and acceptance gate evidence require evidence linkage.

Appendix E — Linked integration, security, performance and defect work

Issue	Evidence lead	Outcome qualification
DDP-084	infra#11; infra PR #19	PARTIAL — exact per-scenario browser logs and environment must be checked; hosted PayHere BLOCKED / NOT VERIFIED
DDP-085	backend#91; documents/security/owasp-checklist.pdf	REFERENCED — detailed control findings require independently reviewed checklist
DDP-086	backend#92; documents/testing/performance-	PARTIAL — not complete hosted search-to-

	report.pdf	browser response proof or 99.9% uptime
DDP-092	infra#14	REFERENCED — no claim of zero open defects without final defect register

Hosted PayHere purchase/payment completion stays BLOCKED / NOT VERIFIED. Formal staff usability and hosted 99.9% uptime are also not verified. NFR-001 p95 backend threshold evidence reported for 13 selected local endpoints does not independently prove the SRS browser-display criterion. Any remaining unverified item must be tracked transparently rather than marked PASS.

Appendix F — QA sign-off judgment

SUPPORTED: local automated suite pass status and aggregate code coverage at historical observations. NOT SUPPORTED YET: a complete API endpoint integration pass rate, release-SHA parity, complete CI artifact trace, every critical business module at 90% branches, and hosted PayHere processing. Final QA sign-off remains CONDITIONAL/PENDING until those checks are evidenced. This document includes available assessments and clearly identified residual verification tasks, rather than invented outcomes.